

“YET IT SEEMS **NOT TO REST UPON SOLID MASONRY**, BUT TO COVER THE SPACE WITH ITS GOLDEN DOME **SUSPENDED FROM HEAVEN.**”

Procopius, Byzantine scholar, from *The Buildings Book*, c.500–65

MUCH ANCIENT KNOWLEDGE

reached the Middle Ages through the efforts of Roman nobleman **Boethius** (c.480–c.524). He acted as a link in the **transmission of Greek and Roman science** to scholars of his time. He translated sections of Aristotle's *Logic*, produced an adaption of Greco-Roman mathematician Nicomachus's (c.60–c.120) *Arithmetike Eisagoge* (*Introduction to Arithmetic*), and **compiled manuals of the liberal arts**, including accounts of Euclidean

geometry and Ptolemaic astronomy. Without his work, much ancient knowledge might have been lost in western Europe.

Flavius Cassiodorus (c.480–c.575), who succeeded Boethius as the leading Roman nobleman at the court of the Ostrogothic kings of Italy, retired around 540 to a monastery he founded at Vivarium in southern Italy. There, he composed *Institutiones Divinarum et Humanarum Lectionum* (*An Introduction to Divine and Human*

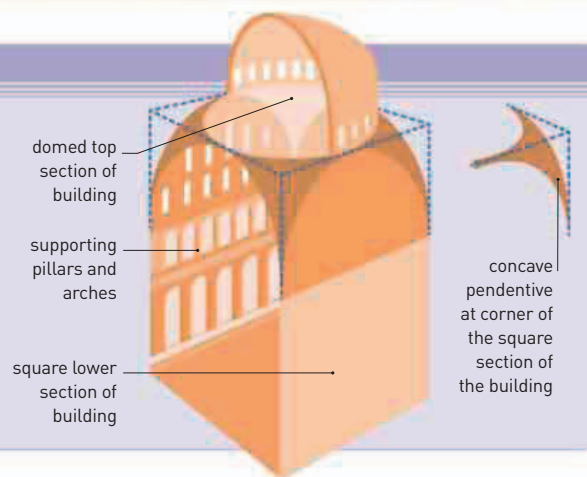
Readings). This handbook on monastic life included a **compilation of secular knowledge**, divided according to the seven liberal arts (see 250–500). Cassiodorus also **established a library** in which many ancient scientific and philosophical

Great minds
Boethius is shown here calculating with written numbers in a competition against Pythagoras, who is using a counting board.



BUILDING WITH PENDENTIVES

Pendentives, such as those employed in the church of Hagia Sophia in Constantinople, are curved, concave sections of masonry that are used to join a square lower section of a building to the circular base of a domed top section. They allow the weight of the dome to be equally distributed onto square supporting walls or piers, which allows far larger domes to be built.



treatises were collected. He instituted the **practice of copying manuscripts**, thus ensuring that important works survived into the later Middle Ages.

Before the 6th century, scholars had largely accepted Aristotle's view that motion was inherent in a body or caused by the medium through which it traveled (such as air). Greek philosopher **John Philoponus** (c.480–c.570) opposed this view, arguing that the medium actually resisted the body's movement. He proposed that **motion is caused externally** through energy impressed upon it by the person or thing moving it. This was the first expression of the theory of **impetus and inertia**.

Around 500, **Li Tao-Yuan** recorded fossil animals in his *Commentary on the Waterways Classic*. He called these fossils

stone oysters, or stone swallows; they were said to emerge from the rock and fly around during thunderstorms. By the mid-7th century, such fossils were being dissolved in vinegar for use as medicine in China.

In early 6th century, Chinese mathematician Zhang Qiujian gave the first example of the **modern method of division**—inverting the divisor and multiplying. He also gave

examples of problems involving **arithmetical progressions** (where the difference between successive terms is constant).

Around 532–37, Byzantine architects **Anthemius of Tralles** (c.474–c.534) and **Isidore of Miletus** succeeded in setting a **round dome over a square room using pendentives**. The dome of Hagia Sophia (in Istanbul, Turkey) remained the largest in the world for nearly a thousand years.



Brachiopod fossil
Resembling bird's wings, these became known as "stone swallows" in China.

- c.500 Cotton gin with worm gearing used in India
- c.500 First use of wooden blocks for printing in China
- c.500 Li Tao-Yuan records fossil brachiopods in China
- 510–24 Boethius writes books on arithmetic and geometry, and translates Aristotle's *Logic*
- 531–79 Nahrwan canal built in Persia
- 532–37 First round dome—in Hagia Sophia church, Turkey—is set over a square room
- c.540 Cosmas Indicopleustes composes *Topographia Christiana*
- c.500 Zhang Qiujian describes the method of division using reciprocal of divisor
- Xu Yue writes *Memoirs on the Mathematical Art*
- 540 Cassiodorus retires to Vivarium and composes *Institutiones Divinarum et Humanarum Lectionum*